Appendix II. Updated Debt Sustainability Analysis Using the Low-Income Country Framework

This appendix provides an annual update to the debt sustainability analysis (DSA) prepared in May 2010.[1] Debt relief under the enhanced Heavily Indebted Poor Countries (HIPC) Initiative and the Multilateral Debt Relief Initiative (MDRI) at the completion point in June 2009 has alleviated C.A.R.'s debt burden and the economy continues to face a moderate risk of debt distress.[2]

1. C.A.R.'s total public debt, including domestic arrears increased markedly, reaching 40 percent of GDP in 2011. This is, well-above the projected level of 27 percent in the 2010 DSA. The debt dynamic in 2010–11 is driven by arrears accumulation on both domestic and external debt and a steady rise in external borrowing from Non-Paris Club members to finance infrastructure projects and social programs. In 2010–11, the C.A.R. contracted concessional loans from India and China for US$52.3 million (2.4 percent of GDP). Accumulation of arrears on external debt amounted US$9.8 million (0.5 percent of GDP)—of which US$1.4 million is due to Paris Club creditors and US$2.6 million to the European Investment Bank, and the balance to other Non-Paris Club creditors that have not yet signed bilateral agreements with C.A.R. Domestic debt accumulation amounted to 2.7 percent of GDP.

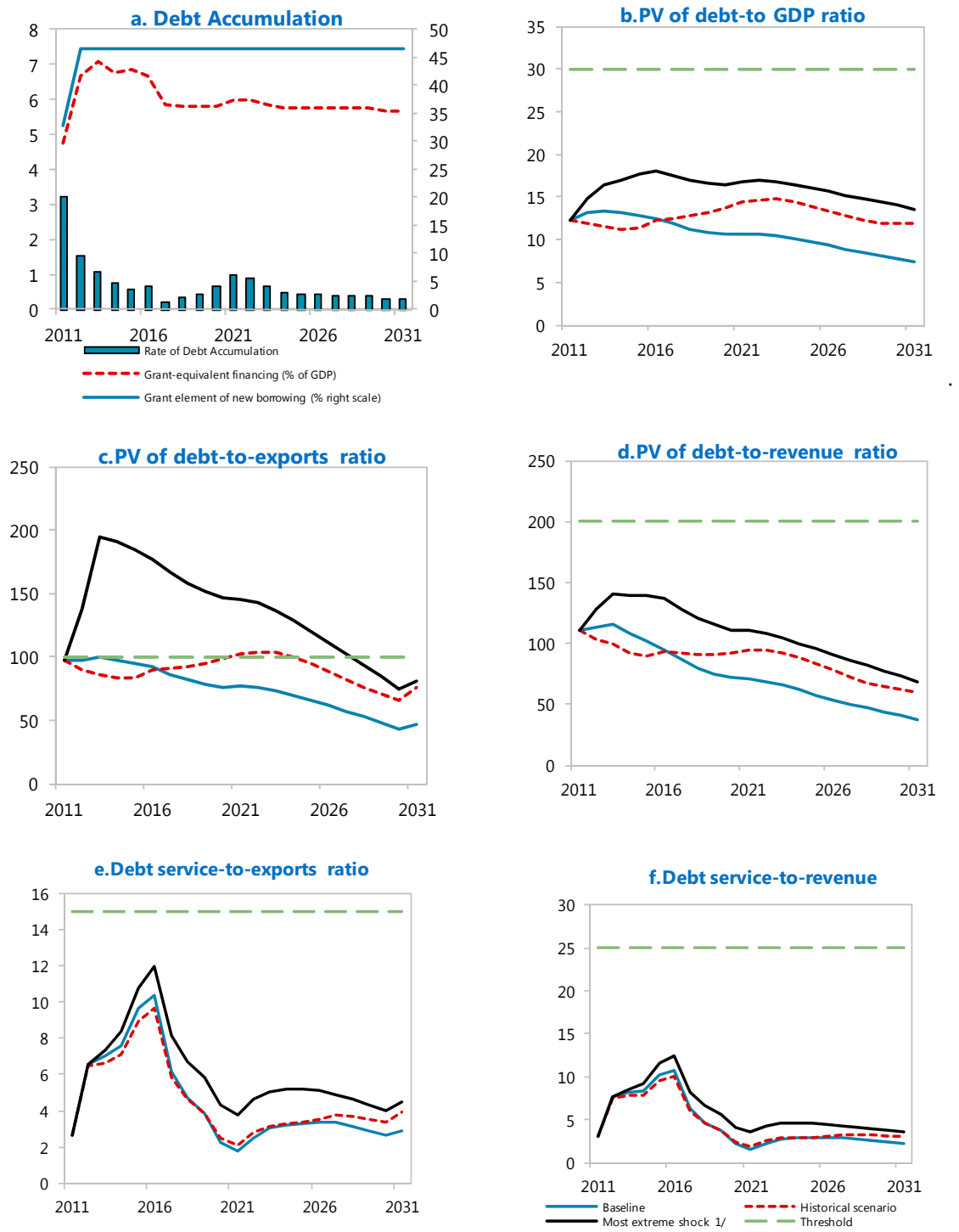
2. The macroeconomic assumptions in the near to medium term for the updated analysis are broadly in line with the previous DSA. Growth is expected to accelerate to around 5 percent per annum on average on the back of scaled-up infrastructures financed by concessional external loans and foreign direct investment (FDI) inflows. The external current account is projected to remain steady as imports related to oil and infrastructure projects would offset rising exports. The fiscal stance is expected to improve on account of increased revenue mobilization and strong budget execution. From 2021 onwards, major mining projects (oil and uranium) would come on stream, boosting growth and government revenue.

3. As in the full DSA, this update suggests that C.A.R. continues to face a "moderate risk of debt distress". Although somewhat higher compared to last year exercise, all baseline debt indicators remain below the thresholds with stable downward trends, but the debt position could still be vulnerable to the most extreme shock to exports, yielding the highest ratio by 2021. Considering the fact that C.A.R. is categorized as having "lower debt vulnerabilities and lower capacity" under the debt sustainability framework, high concessionality is required to preserve debt sustainability.

[1] Full joint Bank-Fund LIC DSAs are prepared once every three years, with short annual updates in the intermediate years (Staff Guidance Note on the Application of the Joint Fund-Bank Debt Sustainability Framework for Low-Income Countries, SM/10/16).

[2] All multilateral creditors with the exception of the Central African Development Bank have provided debt relief. Among Paris Club creditors, agreements have been signed with Austria, Italy, Switzerland, U.S.A., France, Germany, and Japan. For non-Paris Club creditors, agreements have been signed with Saudi Arabia, France Telecom, Assistance Publique des Hôpitaux, and GERBER Company. Negotiations are ongoing with other creditors.

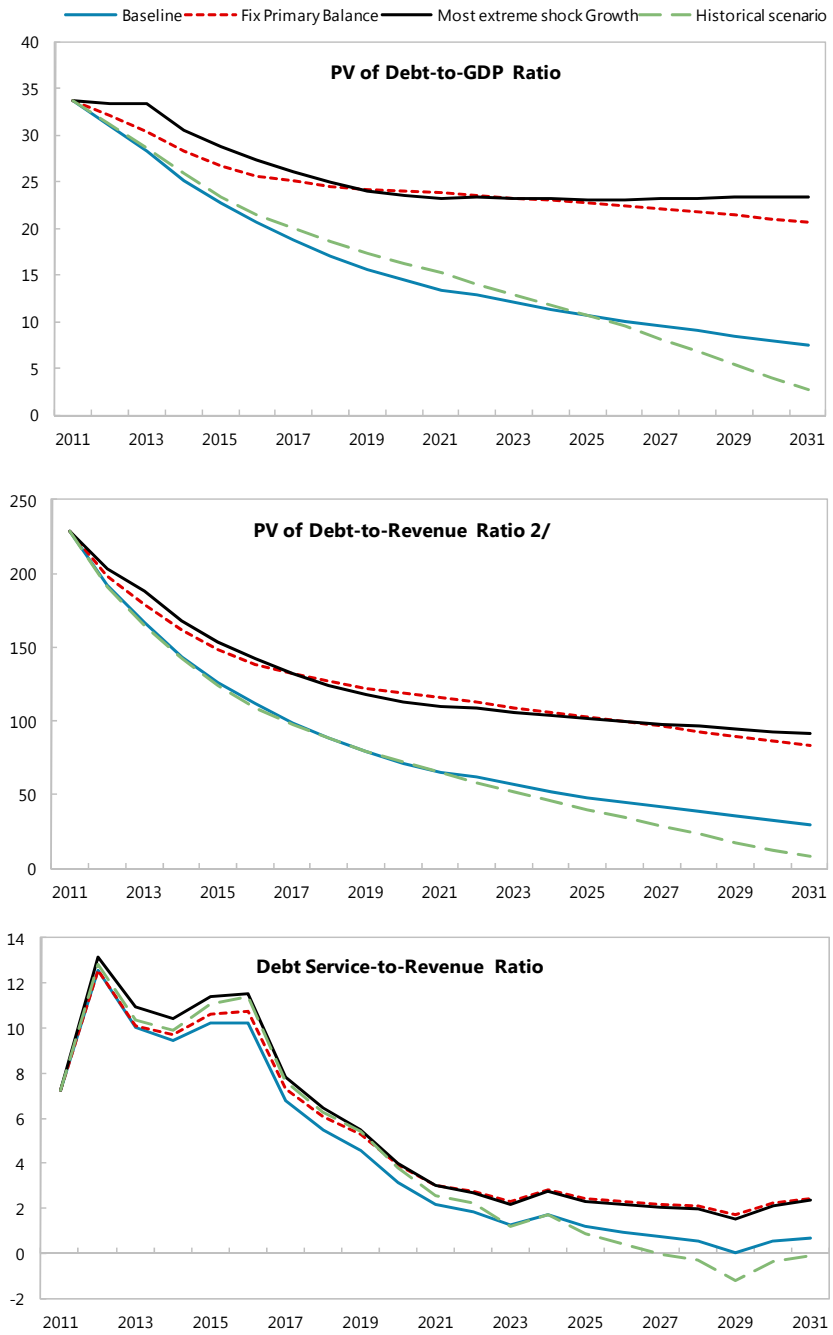
Figure 1. Central African Republic: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2011–31 1/



Sources: C.A.R. authorities; and IMF estimates and projections.

1/ The most extreme stress test is the test that yields the highest ration in 2021. In figure b. it corresponds to a Combination shock; in c. to an Exports shock in d. to a Combination shock; in e. to a Terms shock and in figure f. to a Terms shock.

Figure 2. Central African Republic: Indicators of Public Debt Under Alternative Scenarios, 2011–31 1/



Sources: C.A.R. authorities; and IMF estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2021.
2/ Revenues are defined inclusive of grants.

Table 1a. Central African Republic: External Debt Sustainability Framework, Baseline Scenario, 2008–31 1/

(in percent of GDP, unless otherwise indicated)

	Actual			Historical 2/	Standard 2/	Projections						Projections			
	2008	2009	2010	Average	Deviation	2011	2012	2013	2014	2015	2016	Average 2011-2016	2021	2031	Average 2017-2031
External debt (nominal) 1/	**56.9**	**14.6**	**16.5**			**18.3**	**20.5**	**21.7**	**21.9**	**22.2**	**22.1**		**18.0**	**12.0**	
Of which: Public and publicly guaranteed (PPG)	56.9	14.6	16.5			18.3	20.5	21.7	21.9	22.2	22.1		18.0	12.0	
Change in external debt	2.4	-42.3	1.9			1.8	2.2	1.3	0.2	0.3	-0.1		-0.1	-0.6	
Identified net debt-creating flows	-4.4	6.2	6.8			3.5	3.0	2.0	1.2	0.0	-1.2		-0.3	0.2	
Noninterest current account deficit	**8.9**	**7.7**	**9.7**	**4.4**	**3.6**	**7.2**	**6.8**	**5.9**	**5.3**	**5.1**	**4.5**	**5.8**	**4.5**	**6.2**	**5.4**
Deficit in balance of goods and services	12.6	11.8	13.8			9.8	10.2	10.1	9.5	9.3	8.7		8.6	9.8	
Exports	10.9	9.8	10.8			12.7	13.4	13.5	13.5	13.6	13.7		14.0	15.6	
Imports	23.5	21.6	24.6			22.5	23.6	23.6	23.0	22.9	22.4		22.7	25.5	
Net current transfers (negative = inflow)	-3.8	-3.9	-4.0	-4.0	1.1	-2.7	-3.5	-4.2	-4.2	-4.2	-4.1	-3.8	-4.0	-3.5	-3.9
Of which: Official	-3.6	-3.6	-3.7			-2.3	-3.1	-3.9	-3.9	-3.9	-3.9		-3.9	-3.6	
Other current account flows (negative = net inflow)	0.1	-0.2	0.0			0.1	0.1	0.0	0.0	0.0	-0.1		-0.1	-0.1	
Net FDI (negative = inflow)	**-6.2**	**-2.1**	**-3.1**	**-2.5**	**1.6**	**-3.2**	**-3.2**	**-3.1**	**-3.0**	**-4.1**	**-4.7**	**-3.5**	**-3.9**	**-5.3**	**-4.2**
Endogenous debt dynamics 3/	**-7.1**	**0.7**	**0.2**			**-0.4**	**-0.6**	**-0.8**	**-1.1**	**-1.0**	**-1.1**		**-0.9**	**-0.7**	
Contribution from nominal interest rate	0.9	0.5	0.2			0.0	0.0	0.1	0.1	0.1	0.1		0.1	0.1	
Contribution from real GDP growth	-0.9	-1.0	-0.5			-0.5	-0.7	-0.8	-1.2	-1.1	-1.2		-1.0	-0.8	
Contribution from price and exchange rate changes	-7.1	1.2	0.5			…	…	…	…	…	…		…	…	
Residual (3-4) 4/	**6.8**	**-48.5**	**-4.9**			**-1.7**	**-0.8**	**-0.7**	**-0.9**	**0.2**	**1.1**		**0.2**	**-0.8**	
Of which: Exceptional financing	-1.9	-1.7	0.0			0.5	0.4	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 5/	...	...	10.7			12.3	13.1	13.4	13.2	12.9	12.5		10.7	7.4	
In percent of exports	...	...	99.3			97.0	97.7	99.4	97.8	94.9	91.6		76.4	47.2	
PV of PPG external debt	**...**	**...**	**10.7**			**12.3**	**13.1**	**13.4**	**13.2**	**12.9**	**12.5**		**10.7**	**7.4**	
In percent of exports	**...**	**...**	**99.3**			**97.0**	**97.7**	**99.4**	**97.8**	**94.9**	**91.6**		**76.4**	**47.2**	
In percent of government revenues	**...**	**...**	**92.2**			**110.4**	**113.2**	**115.6**	**108.0**	**101.5**	**94.9**		**70.8**	**37.5**	
Debt service-to-exports ratio (in percent)	**-2.4**	**-2.7**	**0.6**			**2.7**	**6.6**	**7.0**	**7.6**	**9.6**	**10.4**		**1.8**	**2.9**	
PPG debt service-to-exports ratio (in percent)	**-2.4**	**-2.7**	**0.6**			**2.7**	**6.6**	**7.0**	**7.6**	**9.6**	**10.4**		**1.8**	**2.9**	
PPG debt service-to-revenue ratio (in percent)	**-2.5**	**-2.5**	**0.6**			**3.0**	**7.6**	**8.2**	**8.4**	**10.3**	**10.8**		**1.6**	**2.3**	
Total gross financing need (Billions of U.S. dollars)	0.0	0.1	0.1			0.1	0.1	0.1	0.1	0.1	0.0		0.0	0.1	
Noninterest current account deficit that stabilizes debt ratio	6.5	50.0	7.9			5.4	4.6	4.6	5.1	4.9	4.6		4.6	6.8	
Key macroeconomic assumptions															
Real GDP growth (in percent)	2.0	1.7	3.3	1.1	3.2	3.1	4.1	4.2	5.9	5.4	5.7	4.7	6.0	6.9	6.2
GDP deflator in U.S. dollar terms (change in percent)	14.9	-2.0	-3.1	7.2	8.1	8.4	3.0	1.2	1.6	1.4	2.2	3.0	2.3	2.3	2.2
Effective interest rate (percent) 6/	2.0	0.8	1.3	1.0	0.5	0.1	0.3	0.3	0.3	0.3	0.3	0.3	0.5	0.7	0.6
Growth of exports of G&S (U.S. dollar terms, in percent)	-9.8	-10.1	10.3	1.9	12.5	31.6	13.0	6.2	7.4	7.7	8.9	12.5	9.1	-5.0	9.7
Growth of imports of G&S (U.S. dollar terms, in percent)	17.2	-8.4	14.3	8.5	13.0	2.1	12.5	5.3	4.9	6.5	5.7	6.2	9.9	0.0	9.6
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	32.8	46.5	46.5	46.5	46.5	46.5	44.2	46.5	46.5	46.5
Government revenues (excluding grants, in percent of GDP)	10.4	10.8	11.6			11.2	11.6	11.6	12.2	12.7	13.2		15.1	19.7	16.5
Aid flows (in billions of U.S. dollars) 7/	0.1	0.1	0.1			0.1	0.2	0.2	0.2	0.2	0.3		0.3	0.7	
Of which: Grants	0.1	0.1	0.1			0.1	0.1	0.1	0.1	0.2	0.2		0.2	0.6	
Of which: Concessional loans	0.0	0.0	0.0			0.0	0.1	0.1	0.1	0.1	0.1		0.1	0.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			4.7	6.7	7.1	6.8	6.9	6.7		6.0	5.7	5.8
Grant-equivalent financing (in percent of external financing) 8	...	...	...			66.0	73.7	78.4	80.1	79.6	81.1		88.5	92.7	91.1
Memorandum items:															
Nominal GDP (billions of U.S. dollars)	2.0	2.0	2.0			2.2	2.4	2.5	2.7	2.9	3.1		4.6	10.8	
Nominal dollar GDP growth	17.2	-0.3	0.1			11.8	7.2	5.5	7.6	6.9	8.0	7.8	8.3	9.3	8.6
PV of PPG external debt (in billions of U.S. dollars)			0.2			0.3	0.3	0.3	0.4	0.4	0.4		0.5	0.8	
(PVt-PVt-1)/GDPt-1 (in percent)						3.2	1.5	1.1	0.8	0.6	0.7	1.3	1.0	0.3	0.5
Gross workers' remittances (billions of U.S. dollars)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	10.8			12.4	13.2	13.5	13.2	12.9	12.6		10.8	7.4	
PV of PPG external debt (in percent of exports + remittances)	...	...	104.3			101.1	101.5	103.2	101.5	98.3	94.8		78.4	47.8	
Debt service of PPG external debt (in percent of exports + rem	...	...	0.7			2.8	6.8	7.3	7.9	10.0	10.8		1.8	2.9	

Sources: Central African Republic authorities; and IMF estimates and projections.

1/ Includes both public and private sector external debt.
2/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
3/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
4/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments.
For projections also includes contribution from price and exchange rate changes.
5/ Assumes that PV of private sector debt is equivalent to its face value.
6/ Current-year interest payments divided by previous period debt stock.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b. Central African Republic: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2011–31

(in percent)

							Projections										
	2011	2012	2013	2014	2015	2016	**2021**	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of debt-to GDP ratio																	
Baseline	12	13	13	13	12	11	**8**	8	8	7	7	7	6	6	6	5	5
A. Alternative scenarios																	
A1. Key variables at their historical averages in 2011-2031 1/	12	12	12	11	11	11	**12**	12	12	12	12	11	10	10	10	10	10
A2. New public sector loans on less favorable terms in 2011-2031 2	12	15	16	17	16	15	**13**	13	12	12	11	11	11	10	10	10	10
B. Bound tests																	
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	12	14	15	15	14	12	**9**	9	9	8	8	7	7	7	6	6	6
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	12	15	17	17	16	14	**11**	11	10	10	9	8	8	7	7	7	6
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2012-2013	12	14	14	14	13	11	**9**	9	8	8	7	7	7	6	6	6	5
B4. Net nondebt creating flows at historical average minus one standard deviation in 2012-2013 4/	12	15	17	17	15	14	**11**	11	10	9	9	8	8	7	7	6	6
B5. Combination of B1-B4 using one-half standard deviation shocks	12	16	20	20	18	17	**13**	13	12	11	10	10	9	8	8	7	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	12	18	19	18	17	15	**12**	11	11	10	10	9	9	8	8	7	7
PV of debt-to-exports ratio																	
Baseline	97	97	99	97	89	78	**60**	57	54	51	47	43	40	36	33	29	33
A. Alternative scenarios																	
A1. Key variables at their historical averages in 2011-2031 1/	97	89	86	83	79	79	**88**	88	87	84	79	73	67	62	57	54	63
A2. New public sector loans on less favorable terms in 2011-2031 2	97	110	120	126	121	110	**92**	89	86	82	78	73	69	64	59	54	62
B. Bound tests																	
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	97	97	99	97	89	78	**60**	57	54	51	47	43	40	36	33	29	33
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	97	138	194	190	176	157	**121**	115	108	101	92	84	76	69	61	54	60
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2012-2013	97	97	99	97	89	78	**60**	57	54	51	47	43	40	36	33	29	33
B4. Net nondebt creating flows at historical average minus one standard deviation in 2012-2013 4/	97	109	126	123	114	102	**78**	75	70	65	60	55	50	45	40	35	39
B5. Combination of B1-B4 using one-half standard deviation shocks	97	134	183	178	165	148	**114**	109	102	95	87	79	71	64	57	50	55
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	97	97	99	97	89	78	**60**	57	54	51	47	43	40	36	33	29	33
PV of debt-to-revenue ratio																	
Baseline	110	113	115	108	95	81	**55**	52	48	45	41	38	35	32	30	28	26
A. Alternative scenarios																	
A1. Key variables at their historical averages in 2011-2031 1/	110	103	100	92	85	82	**81**	80	78	74	69	64	59	55	52	51	50
A2. New public sector loans on less favorable terms in 2011-2031 2	110	128	140	139	129	114	**85**	81	77	73	68	64	61	57	54	51	49
B. Bound tests																	
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	110	119	130	121	107	91	**62**	59	55	51	47	43	39	36	34	31	30
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	110	126	150	140	125	108	**74**	70	65	59	54	49	45	41	37	34	32
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2012-2013	110	117	122	114	101	86	**59**	55	51	47	44	40	37	34	32	29	28
B4. Net nondebt creating flows at historical average minus one standard deviation in 2012-2013 4/	110	127	147	136	122	105	**72**	68	63	57	52	48	44	40	36	33	31
B5. Combination of B1-B4 using one-half standard deviation shocks	110	137	174	161	145	126	**87**	81	75	68	62	56	51	47	42	39	36
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	110	158	162	151	134	114	**78**	73	68	63	58	53	49	45	42	39	37

Table 1c. Central African Republic: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2011–31 (concluded)

(in percent)

Debt service-to-exports ratio																	
Baseline	3	7	7	8	10	10	**1**	2	3	3	3	3	3	2	2	2	2
A. Alternative scenarios																	
A1. Key variables at their historical averages in 2011-2031 1/	3	6	7	7	9	10	**2**	3	3	3	3	3	3	3	3	3	3
A2. New public sector loans on less favorable terms in 2011-2031 2	3	7	7	8	11	12	**3**	4	4	4	4	4	4	3	3	3	3
B. Bound tests																	
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	3	7	7	8	10	10	**1**	2	3	3	3	3	3	2	2	2	2
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	3	8	11	12	15	16	**3**	4	5	6	6	6	5	5	4	4	4
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2012-2013	3	7	7	8	10	10	**1**	2	3	3	3	3	3	2	2	2	2
B4. Net nondebt creating flows at historical average minus one standard deviation in 2012-2013 4/	3	7	7	8	10	11	**2**	2	3	4	4	4	3	3	3	3	3
B5. Combination of B1-B4 using one-half standard deviation shocks	3	8	10	11	13	14	**2**	3	5	6	6	5	5	4	4	4	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	3	7	7	8	10	10	**1**	2	3	3	3	3	3	2	2	2	2
Debt service-to-revenue ratio																	
Baseline	3	8	8	8	10	11	**1**	2	2	3	3	3	2	2	2	2	2
A. Alternative scenarios																	
A1. Key variables at their historical averages in 2011-2031 1/	3	8	8	8	9	10	**2**	2	3	3	3	3	3	3	3	3	3
A2. New public sector loans on less favorable terms in 2011-2031 2	3	8	9	9	12	12	**3**	3	4	4	4	3	3	3	3	3	3
B. Bound tests																	
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	3	8	9	9	12	12	**2**	2	3	3	3	3	3	3	2	2	2
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	3	8	8	9	11	11	**2**	2	3	4	3	3	3	3	3	2	2
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2012-2013	3	8	9	9	11	11	**1**	2	3	3	3	3	3	2	2	2	2
B4. Net nondebt creating flows at historical average minus one standard deviation in 2012-2013 4/	3	8	8	9	11	11	**2**	2	3	3	3	3	3	3	3	2	2
B5. Combination of B1-B4 using one-half standard deviation shocks	3	8	9	10	12	12	**2**	3	3	4	4	4	4	3	3	3	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	3	11	12	12	14	15	**2**	3	3	4	4	4	3	3	3	3	3
Memorandum item:																	
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	43	43	43	43	43	43	**43**	43	43	43	43	43	43	43	43	43	43

Sources: Country authorities; and IMF estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock import levels (implicitly assuming an offsetting adjustment in).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a. Central African Republic: Public Sector Debt Sustainability Framework, Baseline Scenario, 2008–31

(in percent of GDP, unless otherwise indicated)

	Actual			Standard		Estimate	Projections								
	2008	2009	2010			2011	2012	2013	2014	2015	2016	Average 2011–16	2021	2031	Average 2017–31
Public sector debt 2/	80.3	35.0	37.8			39.6	38.3	36.6	33.8	32.0	30.2		20.7	12.1	
Of which: Foreign-currency denominated	56.9	14.6	16.5			18.3	20.5	21.7	21.9	22.2	22.1		18.0	12.0	
Change in public sector debt	1.2	-45.3	2.8			1.8	-1.3	-1.7	-2.8	-1.7	-1.8		-1.2	-0.7	
Identified debt-creating flows	-4.0	-8.2	1.3			0.4	-1.5	-1.1	-2.3	-1.2	-1.7		-2.3	-0.9	
Primary deficit	-0.9	-1.0	0.4	-0.8	3.6	2.3	0.5	0.5	0.0	0.7	0.4	0.7	0.0	0.5	0.3
Revenue and grants	15.2	16.1	18.0			14.7	16.2	17.0	17.5	18.0	18.5		20.5	24.9	
Of which: grants	4.8	5.3	6.3			3.5	4.6	5.4	5.3	5.3	5.3		5.3	5.3	
Primary (noninterest) expenditure	14.3	15.1	18.4			17.0	16.6	17.5	17.6	18.7	18.9		20.5	25.4	
Automatic debt dynamics	-1.2	-6.6	0.8			-2.1	-1.7	-1.6	-2.3	-1.9	-2.1		-1.5	-1.1	
Contribution from interest rate/growth differential	-4.8	-2.9	-0.7			-0.8	-1.9	-1.8	-2.5	-2.1	-2.1		-1.5	-1.1	
Of which: Contribution from average real interest rate	-3.2	-1.6	0.4			0.4	-0.4	-0.3	-0.5	-0.4	-0.4		-0.3	-0.3	
Of which: Contribution from real GDP growth	-1.6	-1.3	-1.1			-1.2	-1.5	-1.6	-2.0	-1.7	-1.7		-1.2	-0.8	
Contribution from real exchange rate depreciation	3.6	-3.6	1.5			-1.3	0.2	0.2	0.2	0.2	0.0		...	...	
Other identified debt-creating flows	-1.9	-0.7	0.1			0.2	-0.3	0.0	0.0	0.0	0.0		-0.8	-0.3	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-1.9	-0.7	0.1			0.2	-0.3	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g., bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		-0.8	-0.3	
Residual, including asset changes	5.1	-37.0	1.5			1.5	0.3	-0.6	-0.5	-0.5	-0.1		1.1	0.2	
Other Sustainability Indicators															
PV of public sector debt	...	...	32.0			33.6	31.0	28.3	25.0	22.7	20.7		13.4	7.4	
Of which: Foreign-currency denominated	...	...	10.7			12.3	13.1	13.4	13.2	12.9	12.5		10.7	7.4	
Of which: External	...	...	10.7			12.3	13.1	13.4	13.2	12.9	12.5		10.7	7.4	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 3/	-0.3	-0.6	1.3			3.4	2.5	2.2	1.7	2.6	2.3		0.5	0.7	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	…	177.9			228.4	191.8	166.7	143.0	126.1	111.6		65.4	29.9	
PV of public sector debt-to-revenue ratio (in percent)	…	…	274.8			300.9	267.9	244.0	205.2	179.2	156.7		88.3	37.9	
Of which: External 4/	…	…	92.2			110.4	113.2	115.6	108.0	101.5	94.9		70.8	37.5	
Debt service-to-revenue and grants ratio (in percent) 5/	4.0	2.6	4.9			7.3	12.6	10.0	9.4	10.2	10.3		2.2	0.7	
Debt service-to-revenue ratio (in percent) 5/	5.8	3.9	7.5			9.6	17.6	14.7	13.5	14.5	14.4		2.9	0.8	
Primary deficit that stabilizes the debt-to-GDP ratio	-2.0	44.3	-2.4			0.5	1.7	2.2	2.9	2.5	2.2		1.2	1.2	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	2.0	1.7	3.3	1.1	3.2	3.1	4.1	4.2	5.9	5.4	5.7	4.7	6.0	6.9	6.2
Average nominal interest rate on forex debt (in percent)	2.0	0.8	1.3	1.0	0.5	0.1	0.3	0.3	0.3	0.3	0.3	0.3	0.5	0.7	0.6
Average real interest rate on domestic debt (in percent)	-3.0	-0.4	2.4	-0.5	2.4	2.3	0.2	0.5	-0.1	0.1	0.1	0.5	-0.3	-52.4	-6.9
Real exchange rate depreciation (in percent, + indicates depreciation)	7.0	-6.7	10.4	-3.3	10.4	-8.2	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	7.0	3.6	1.7	3.2	1.9	0.8	2.8	2.1	2.6	2.2	2.2	2.1	2.3	2.3	2.2
Growth of real primary spending (deflated by GDP deflator, in percent)	0.2	0.1	0.3	0.1	0.2	0.0	0.0	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Grant element of new external borrowing (in percent)	...	...	...	…	…	32.8	46.5	46.5	46.5	46.5	46.5	44.2	46.5	46.5	...

Sources: Central African Republic authorities; and IMF estimates and projections.

1/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
2/ Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.
3/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
4/ Revenues excluding grants.
5/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

Table 2b. Central African Republic: Sensitivity Analysis for Key Indicators of Public Debt 2011–31

	Projections							
	2011	2012	2013	2014	2015	2016	2021	2031
PV of Debt-to-GDP Ratio								
Baseline	34	31	28	25	22	19	13	6
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	34	31	29	26	23	21	16	4
A2. Primary balance is unchanged from 2011	34	32	30	28	26	25	24	21
A3. Permanently lower GDP growth 1/	34	31	29	26	23	20	16	18
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	34	33	33	30	28	25	22	22
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	34	32	31	28	24	21	14	7
B3. Combination of B1-B2 using one half standard deviation shocks	34	33	32	29	26	23	19	17
B4. One-time 30 percent real depreciation in 2012	34	35	32	28	24	21	13	6
B5. 10 percent of GDP increase in other debt-creating flows in 2012	34	37	34	30	27	23	15	8
PV of Debt-to-Revenue Ratio 2/								
Baseline	228	191	166	143	121	103	61	24
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	228	191	165	143	122	106	68	13
A2. Primary balance is unchanged from 2011	228	198	179	161	147	136	118	86
A3. Permanently lower GDP growth 1/	228	193	169	146	126	109	77	69
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	228	202	188	167	148	132	105	85
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	228	200	182	157	135	115	70	28
B3. Combination of B1-B2 using one half standard deviation shocks	228	200	180	159	139	123	92	68
B4. One-time 30 percent real depreciation in 2012	228	218	186	157	132	112	66	24
B5. 10 percent of GDP increase in other debt-creating flows in 2012	228	227	199	173	147	124	75	33
Debt Service-to-Revenue Ratio 2/								
Baseline	7	13	10	9	10	11	6	1
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	7	13	10	10	11	12	7	1
A2. Primary balance is unchanged from 2011	7	13	10	10	11	11	7	3
A3. Permanently lower GDP growth 1/	7	13	10	10	11	11	6	2
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	7	13	11	10	12	12	7	3
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	7	13	10	10	11	11	6	1
B3. Combination of B1-B2 using one half standard deviation shocks	7	13	11	10	11	12	7	2
B4. One-time 30 percent real depreciation in 2012	7	14	12	12	14	14	6	2
B5. 10 percent of GDP increase in other debt-creating flows in 2012	7	13	11	10	11	11	6	2

Sources: Central African Republic authorities; and IMF estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.